

Metadata

Dataset Title

Short name or nickname you use to refer to this dataset: NPS Visitation

Abstract

This map is supposed to help inform viewers of the most visited National Parks in the United States. The intended audience is anyone who is planning a trip to a National Park or who wants to review visual data about them. If I were planning a trip to a National Park, I would use this map to pick parks that have less visitation. Throughout this project, I had to do various functions on ArcGISPro, including geoprocessing tools, and then coding on R Studio. Hope you enjoy!

Investigators

(list in order as for a paper with e-mail addresses, organization and preferably ORCID ID, if you don't have one, get it, it's easy and free: <http://orcid.org/>) add table rows as needed

First Name	Middle Initial	Last Name	Organization	e-mail address	ORCID ID (optional)
Cassadie	S	Mefferd	Colorado Mountain College	csmefferd@mail.colradomtn.edu	

Other personnel names and roles

(dataset creators & contact, field crew, data entry etc. with e-mail addresses, organization and ORCID ID)

First Name	Middle Initial	Last Name	Organization	e-mail address	ORCID ID (optional)	Role in project

License

N/A

Keywords

National Parks, NPS, United States, Visitation, Outdoor Recreation, Busiest, Nature, Environment, Trip Planning

Funding of this work:

Add rows to table if several grants were involved, list only the main PI, start with main grant first:

PI First Name	PI Middle Initial	PI Last Name	PI ORCID ID (optional)	Title of Grant	Funding Agency	Funding Identification Number

Timeframe

- Begin date: 4/24/25
- End date: 4/29/25
- Data collection ongoing/completed: Completed

Geographic location

- Verbal description: United States
- North bounding coordinate (decimal degree): 71.439786
- South bounding coordinate (decimal degree): -14.601813
- East bounding coordinate (decimal degree): 146.154418
- West bounding coordinate (decimal degree): -178.443593

Taxonomic species or groups

N/A

Methods

Make a new map on ArcGISPro

Insert nps_boundary shape file

Geoprocessing tool = (Project) to new coordinate system (North American albers equal area conic)

Then we want to turn the boundaries into points to make it look better for the maps intention

Geoprocessing tool = (feature to point)

Total in attribute table is 435, so we want to sort out just the national parks. So select by attribute [where "UNIT_TYPE" is equal to "National Park"]

Make layer from selected features and boom, narrowed it down to 60

Now we can add the "tl_2023_us_state.shp"

And we need to project this layer to match the other one in (North American albers equal area conic) by using the project tool again

Now lets clip the nps points to the usa outline just to make sure we dont have any scaglers

Switch over to R Studio to find average visits for the park

Turn the US-National-Parks-RecreationVisits_1979-2024.csv into something we can use.

1. Using R Studio

```
> setwd("D:\\Cassadie_Mefferd\\Cartography_Final_Project")
```

install.packages("tidyverse") to get:

```
read_csv() for reading in your data
```

dplyr for data wrangling (like finding averages)

ggplot2 for graphing

tidyr for reshaping if needed

#to read the data

```
visits <- read_csv("US-National-Parks_RecreationVisits_1979-2024.csv")
```

```
#to get an average of the visitation rates per year for each park
```

```
avg_visits <- visits %>%
```

```
  group_by(ParkName) %>%
```

```
  summarize(AverageVisits = mean(RecreationVisits, na.rm = TRUE))
```

```
#to download the average visitation rates as a file
```

```
write_csv(avg_visits, "average_visits_by_park.csv")
```

Back in ArcGISPro, join average data to the nps points. But it says null in every column, so we need to clean up the names of the parks in each table to that they match, we can do that on R

The points names of the national parks dont match the csv data

```
> average_visits_clean$ParkName <- gsub(" NP$", "",
average_visits_clean$ParkName)
> write_csv(average_visits_clean, "average_visits_clean_noNP.csv")

> average_visits_clean$ParkName <- gsub(" NP$", "",
average_visits_clean$ParkName)
> head(average_visits_clean$ParkName)
[1] "Acadia Np" "Arches
Np"
[3] "Badlands Np" "Big Bend
Np"
[5] "Biscayne Np" "Black Canyon Of The Gunnison
Np"
> average_visits_clean$ParkName <- gsub(" NP", "",
average_visits_clean$ParkName, ignore.case = TRUE)
>
> average_visits_clean$ParkName <-
trimws(average_visits_clean$ParkName)
> head(average_visits_clean$ParkName)
[1] "Acadia" "Arches"
[3] "Badlands" "Big Bend"
[5] "Biscayne" "Black Canyon Of The Gunnison"

#to save
> write_csv(average_visits_clean, "average_visits_clean_noNP.csv")
```

Whoohoo it finally worked, now go back into arc and join the csv with the points.

Now we get to make it show up on a map and work on making it pretty, so we will symbolize!
Choose graduated colors

When I chose equal interval, the numbers were overwhelming, so I changed it to manual interval and set 5 classes to be a range of 2,000,000 average visitors annually.

I am trying to decide between graduated symbols and proportional symbols. I decided to go with proportional symbols because it shows bigger symbols for the higher trafficked parks, and it represents each data point by its value, not just a range. So I also added a green color scheme, so the highest visitation is dark green and lowest is light green. I also made the circles slightly transparent so that you could see any overlapping, or the background layer.

Then I turned on labels and now its time to mess with those.

Download as a map package and open in Q gis to mess with labels.

Well I couldnt figure that out, so im messing with labels in arc

Make a layout, add legend, title, scale bar, north arrow and make it look pretty.

Data Table

Table name: NPS_Data.csv

Table description: All of the National Parks in the US, what state they are in, and average visitation.

Column name	Description	Unit or code explanation or date format	Empty value code

unit_code	park_name	state	AvgVisits
ACAD	Acadia	ME	3001627.152
ARCH	Arches	UT	894382.8913
BADL	Badlands	SD	1022875.587
BIBE	Big Bend	TX	326903.3478
BISC	Biscayne	FL	463649
BLCA	Black Canyon of the Gunnison	CO	245845.5435
BRCA	Bryce Canyon	UT	1233760.565
CANY	Canyonlands	UT	419784.9783
CARE	Capitol Reef	UT	682313.7174
CAVE	Carlsbad Caverns	NM	538724.4783
CHIS	Channel Islands	CA	312260.2826
CONG	Congaree	SC	103145.775
CRLA	Crater Lake	OR	486658.5
CUVA	Cuyahoga Valley	OH	2211137.696
DENA	Denali	AK	431212.6739
DEVA	Death Valley	CA	948566.9348
DRTO	Dry Tortugas	FL	47996.5
EVER	Everglades	FL	922629.7391
GAAR	Gates of the Arctic	AK	6970.744186
GLAC	Glacier	MT	2095725.435
GLBA	Glacier Bay	AK	334073.7826
GRBA	Great Basin	NV	86826.69565
GRCA	Grand Canyon	AZ	4161919.826
GRSA	Great Sand Dunes	CO	298985.5217
GRSM	Great Smoky Mountains	NC	9822658.326
GRTE	Grand Teton	WY	2509523.935
GUMO	Guadalupe Mountains	TX	182275.9783
HALE	Haleakala	HI	1179257.522
HAVO	Hawaii Volcanoes	HI	1411694.696
HOSP	Hot Springs	AR	1432560.804

INDU	Indiana Dunes	IN	1871785.652
ISRO	Isle Royale	MI	20314.13043
JEFF	Gateway Arch	IL	2419498.652
JOTR	Joshua Tree	CA	1482932.804
KATM	Katmai	AK	40159.84444
KEFJ	Kenai Fjords	AK	227288.5581
KICA	Kings Canyon	CA	685076.3043
KOVA	Kobuk Valley	AK	6450.162791
LACL	Lake Clark	AK	11510.02326
LAVO	Lassen Volcanic	CA	418994.1087
MACA	Mammoth Cave	KY	1281422.609
MEVE	Mesa Verde	CO	565156.8913
MORA	Mount Rainier	WA	1307281.891
NOCA	North Cascades	WA	195927.8478
OLYM	Olympic	WA	3003451.609
PEFO	Petrified Forest	AZ	694281.1087
PINN	Pinnacles	CA	200763.7609
REDW	Redwood	CA	448139.0217
ROMO	Rocky Mountain	CO	3137569.196
SAGU	Saguaro	AZ	729941.7391
SEQU	Sequoia	CA	1009114.478
SHEN	Shenandoah	VA	1524167.304
THRO	Theodore Roosevelt	ND	536510.0435
VOYA	Voyageurs	MN	221786.913
WHSa	White Sands	NM	549837.4348
WICA	Wind Cave	SD	611498.5217
WRST	Wrangell - St Elias	AK	49340.62791
YELL	Yellowstone	ID	3133457.217
YOSE	Yosemite	CA	3474583.13
ZION	Zion	UT	2697031.913

Articles

Article DOI or URL (DOI is preferred)	Article title	Journal title

Scripts/code (software)

File name	Description	Scripting language
		R

#to create a histogram with R Studio

```
nps_data=read.csv("D:/Cassadie_Mefferd/nps_data.csv")
```

#to get rid of scientific notation

```
options(scipen = 999)
```

#this makes the actual hisogram

```
hist(nps_data$AvgVisits,main = "Average National Park Visits",  
     xlab = "Average Visits",xlim = c(0,10000000), breaks=seq(0,10000000,100000),  
     col = "forestgreen")
```

#to get rid of Great Smokey Mountain NP Outlier

```
nps_data_noGSM <- subset(nps_data, AvgVisits < 5000000)  
hist(nps_data_noGSM$AvgVisits,main = "Average National Park Visits",  
     xlab = "Average Visits",xlim = c(0,5000000), breaks=seq(0,5000000,50000),  
     col = "forestgreen")
```

Data provenance

Dataset title	Dataset DOI or URL	Creator (name & ...)	Contact (name & email)
National Park Service boundaries	https://doi.org/10.21949/1529046	Land Resources Division	Chief Cartographer PPFL_LandsNet@nps.gov
U.S. NATIONAL PARK VISIT DATA (1979-2024)	https://www.responsible-datasets-in-context.com/posts/np-data/	Melanie Walsh and Os Keyes	melwalsh@uw.edu Os_Keyes@uml.edu
TIGER/Line Shapefile, Current, Nation, U.S., State and Equivalent Entities	https://www2.census.gov/geo/tiger/TIGER2023/STATE/tl_2023_us_state.zip	U.S. Department of Commerce, U.S. Census Bureau, Geography Division, Geospatial Products Branch (Point of Contact)	geo.geography@census.gov

Notes and Comments